

输入: N个目标位姿 T^* , K=16 Sobol种子

DA Grid Launch: N blocks, 32 threads/block

Block i: 加载 T^*_i 和 seed[0..15] 到寄存器

16 lanes 并行: 每个lane执行LM迭代求解

LM迭代 (max 50):

1. FK with frames $\rightarrow T, p, z$
2. 解析Jacobian: $Jv = z \times (p - p)$
3. $H = J^T J + \lambda I$, $g = J^T e$
4. 6x6 Gauss Elim
5. 内部迭代

Loss评估:

- Pose cost = $|ep|^2 + |er|^2$
- Limit barrier penalty
- Total = $0.5 \cdot \text{pose} + w \cdot \text{barrier}$
- Trial accept: $q \leftarrow q + dq$

收敛? ($\text{pos} < 5\text{mm}$ & $\text{rot} < 1^\circ$) 或 $\text{iter} = \text{max}$?

No

继续

16个候选解写入Shared Memory $s_cand[]$

Lane 0: 块内候选选择
Success Rank $\rightarrow$ Near-Limit $\rightarrow$ Pose Cost

best[i] 写入Global Memory

输出: $N \times 18$ best [q, pos, rot, cost, rank]

每个Block处理
1个Target的
16个种子

单Kernel
零调度开销
无额外筛选
Kernel Launch